

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Date: 29-08-2026

TASK 1 — HTML AND HTML5

1. TITLE

Develop a College Information Portal Using HTML and HTML5 Features

2. AIM

To understand and implement the fundamental building blocks of **HTML and HTML5** for creating a structured and interactive web page containing text, lists, tables, images, forms, semantic elements, multimedia, and HTML5 input types.

3. OBJECTIVE

The objectives of this task are:

- To understand the basic structure of an HTML5 document.
 - To create web pages using headings and paragraphs.
 - To apply different text-formatting tags. To create ordered and unordered lists.
 - To add hyperlinks to external websites.
 - To create and display student details using tables.
 - To insert images with alternate text.
 - To design a student registration form.
 - To use HTML5 semantic elements.
 - To embed audio and video files.
 - To implement HTML5 input types such as email, date, number, color, and range. To understand the role of HTML in structuring web content.
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4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create and structure web pages. HTML uses elements and tags to define different parts of a web page such as headings, paragraphs, lists, tables, forms, images, and links.

HTML5 is the modern version of HTML that provides additional features such as semantic elements, multimedia support, canvas, and new form input types.

The basic structure of an HTML5 document consists of:

- `<!DOCTYPE html>` — Defines the document as an HTML5 document.
- `<html>` — Root element of the HTML document.
- `<head>` — Contains metadata and the page title.
- `<body>` — Contains the visible content of the web page.

Important concepts used in this task:

Headings and Paragraphs:

`<h1>` to `<h6>` are used for headings, while `<p>` is used for paragraphs.

Text Formatting:

`<b>` makes text bold, `<i>` makes text italic, `<u>` underlines text, `<sup>` displays superscript text, and `<sub>` displays subscript text.

Lists:

`<ul>` creates an unordered list, while `<ol>` creates an ordered list. Individual items are created using `<li>`.

Hyperlinks:

The `<a>` tag is used to create links to other web pages or websites.

Tables:

`<table>`, `<tr>`, `<th>`, and `<td>` are used to create and organize tabular information.

Images:

The `<img>` tag is used to display images. The `alt` attribute provides alternative text if the image cannot be displayed.

Forms:

HTML forms collect user information using controls such as text boxes, radio buttons, checkboxes, dropdowns, textareas, and buttons.

Semantic Elements:

HTML5 provides meaningful elements such as `<header>`, `<nav>`, `<section>`, `<article>`, and `<footer>` to describe the structure of a web page.

Multimedia:

The `<audio>` and `<video>` elements allow audio and video files to be embedded directly into web pages.

HTML5 Input Types:

HTML5 provides specialized input types such as email, date, number, color, and range, which provide better input handling and browser-level validation.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file.
2. Define the document as an HTML5 document using `<!DOCTYPE html>`.
3. Create the basic `<html>`, `<head>`, and `<body>` structure.
4. Add the title **College Information Portal**.
5. Add headings and paragraphs containing college information.
6. Apply bold, italic, underline, superscript, and subscript formatting.
7. Create an unordered list of available courses.
8. Create an ordered list containing the application procedure.
9. Add hyperlinks to the college website and Google.
10. Create a student details table containing Roll No, Name, Branch, and Email.
11. Insert the college logo and campus image using the `<img>` tag.

12. Add appropriate alternate text to the images.
 13. Create a student registration form.
 14. Add name, email, password, gender, hobbies, branch, comments, submit, and reset controls.
 15. Add HTML5 semantic elements such as header, navigation, section, article, and footer.
 16. Embed an audio file using the <audio> element.
 17. Embed a video file using the <video> element.
 18. Add HTML5 input types such as email, date, number, color, and range.
 19. Save the HTML file.
 20. Open the file in a web browser.
 21. Verify all elements, links, images, form controls, audio, and video.
 22. Record the output using screenshots.
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6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>College Information Portal</title>
</head>
<body>
<!--Headings and Paragraphs-->
<h1>College Information Portal</h1>
<h2>About Our Institute</h2>
<p>Welcome to the official Information Portal</p>
<p>Our college provides world-class education and fosters innovation among students</p>
<!--Text Formatting-->
<h2>Academic Excellence</h2>
<p>We are proud to be ranked<b> Number 1</b> in the region. Our<i> State-of-the-art</i>
campus offers a <u>dynamic learning environment</u>.</p>
<p>Check out our award-winning formula :  $E=mc^2$ . Alternatively, look at our
chemistry lab schedules for  $H_2O$  solutions.</p>
<!--Unordered List-->
<h2>Available Courses</h2>
<ul>
<li>Computer Science Engineering</li>
<li>Data Science</li>
<li>Artificial Intelligence</li>
```

```

</ul>
<!--Ordered List-->
<ol>
<li>Fill out the online application form.</li>
<li>Submit your Academic transcript.</li>
<li>Attend the counseling session.</li>
</ol>
<!--HyperLinks-->
<h2>Useful Links</h2>
<p>Visit the official <a href="https://anits.org/" target="_blank">CollegeWebsite</a>for
more details.</p>
<p>For external research, you can search on <a href="https://www.google.com/"
target="_blank">Google</a>.</p>
<h2>Student Details</h2>
<!--Student details table-->
<table border="1">
<tr>
<th>Roll No</th>
<th>Name</th>
<th>Branch</th>
<th>Email</th>
</tr>
<tr>
<td>100</td>
<td>Raj</td>
<td>CSE</td>
<td>raj@gmail.com</td>
</tr>
<tr>
<td>101</td>
<td>Priya</td>
<td>CSM</td>
<td>priya@gmail.com</td>
</tr>
</table>
<!--Images-->
<h2>College Logo</h2>

<h2>Campus Image</h2>

<!--Forms-->
<h2>Student      Registration      Form</h2>
<form>
Name: <input type="text"><br><br>

```

```
Email: <input type="email"><br><br>
Password: <input type="password"><br><br>
Gender: <input type="radio" name="Gender">Male
<input type="radio" name="Gender">Female
<br><br>
Hobbies: <input type="checkbox">Sports
<input type="checkbox">Reading
<input type="checkbox">Music <br><br> Branch:
<select>
<option>CSE</option>
<option>ECE</option>
<option>CSM</option>
<option>CSD</option>
</select> <br><br>
Comments <br>
<textarea rows="4" cols="30"></textarea> <br><br>
<input type="submit">
<input type="reset">
</form>
<!--HTML5 Features-->
<header>
<h1>College Information Portal</h1>
</header>
<nav>
    <a href="#">Home</a>
<a href="#">About</a>
<a href="#">Contact</a>
</nav>
<section>
<article>
<h2>About College</h2>
<p>ANITS college provides quality education.</p>
</article>
</section>
<h3>Audio</h3>
<audio controls>
<source src="audio.mp3" type="audio/mpeg"> </audio>
<h3>Video</h3>
<video width="400"controls>
<source src="video.mp4" type="video/mp4">
</video>
</form>
Email: <input type="email"><br><br>
```

```
Date: <input type="date"><br><br>
Number: <input type="number"><br><br>
Favorite Color: <input type="color"><br><br>
Range: <input type="range"><br><br>
<footer>
<p>Copyright 2026 ANITS College</p>
</footer>
</form>
</body>
</html>
```

7. CODE EXPLANATION

Code / Tag	Explanation
<!DOCTYPE html>	Declares the document as HTML5.
<html>	Defines the root element of the web page.
<head>	Contains metadata and page information.
<title>	Specifies the title displayed in the browser tab.
<h1>, <h2>, <h3>	Define different levels of headings.
<p>	Defines a paragraph.
	Displays text in bold.
<i>	Displays text in italic.
<u>	Underlines text.
<sup>	Displays superscript text.
<sub>	Displays subscript text.
	Creates an unordered list.
	Creates an ordered list.
	Defines an individual list item.
<a>	Creates a hyperlink.
<table>	Creates a table.
<tr>	Defines a table row.
<th>	Defines a table heading.
<td>	Defines table data.
	Displays an image.
alt	Provides alternative text for an image.
<form>	Creates an HTML form.
<input>	Creates different types of input controls.
<select>	Creates a dropdown list.
<textarea>	Creates a multi-line text input area.

<header>	Defines the header section of a page.
<nav>	Defines navigation links.
<section>	Defines a section of content.
<article>	Defines independent or self-contained content.
<footer>	Defines the footer section.
<audio>	Embeds an audio file.
<video>	Embeds a video file.
target="_blank"	Opens the linked page in a new browser tab.

8. TEST CASES AND EXECUTION DATA

The following test cases can be recorded after executing the webpage in a browser.

Test Case	TC-01
Test / Input	Load the College Information Portal webpage in the browser
Expected Result	The webpage should display the college information, available courses, useful links, student details, college logo, campus image, registration form, media elements, and footer correctly.
Actual Result	All the webpage sections and elements were displayed correctly.
Status	Pass

TC-01 Output:

College Information Portal

About Our Institute

Welcome to the official Information Portal

Our college provides world-class education and fosters innovation among students

Academic Excellence

We are proud to be ranked **Number 1** in the region. Our *State-of-the-art* campus offers a

Check out our award-winning formula : $E=mc^2$. Alternatively, look at our chemistry lab

Available Courses

- Computer Science Engineering
- Data Science
- Artificial Intelligence

1. Fill out the online application form.
2. Submit your Academic transcript.
3. Attend the counseling session.

Useful Links

Visit the official [College Website](#) for more details.

For external research, you can search on [Google](#).

Student Details

Roll No	Name	Branch	Email
100	Raj	CSE	raj@gmail.com
101	Priya	CSM	priya@gmail.com

College Logo



Campus Image



Student Registration Form

Name:

Email:

Password:

Gender: ☐ Male ☐ Female

Hobbies: ☐ Sports ☐ Reading ☐ Music

Branch:

Comments:

College Information Portal

[Home](#) [About](#) [Contact](#)

About College

ANITS college provides quality education.

Audio



Video



Email:

Date:

Number:

Favorite Color:

Range:

Copyright 2026 ANITS College

Test Case	TC-02
Test / Input	Check the College Information Portal content and headings
Expected Result	The headings such as College Information Portal, About Our Institute, Academic Excellence, Available Courses, and Useful Links should be displayed correctly.
Actual Result	All headings and corresponding information were displayed correctly.
Status	Pass

TC-02 Output:

College Information Portal

About Our Institute

Welcome to the official Information Portal

Our college provides world-class education and fosters innovation among students

Academic Excellence

We are proud to be ranked **Number 1** in the region. Our *State-of-the-art* campus offers a *dynamic learning environment*

Check out our award-winning formula : $E=mc^2$. Alternatively, look at our chemistry lab schedules for H_2O solutions.

Available Courses

- Computer Science Engineerinh
 - Data Science
 - Artificial Intelligence
1. Fill out the online application form.
 2. Submit your Academic transcript.
 3. Attend the counseling session.

Useful Links

Visit the official [CollegeWebsite](#) for more details.

For external research, you can search on [Google](#).

Test Case	TC-03
Test / Input	Check the Available Courses list
Expected Result	The courses Computer Science Engineering, Data Science, and Artificial Intelligence should be displayed as a list.
Actual Result	All three courses were displayed correctly.
Status	Pass

TC-03 Output:

Available Courses

- Computer Science Engineerinh
 - Data Science
 - Artificial Intelligence
1. Fill out the online application form.
 2. Submit your Academic transcript.
 3. Attend the counseling session.

Test Case	TC-04
Test / Input	Check the Useful Links section
Expected Result	The College Website and Google links should be displayed and available for navigation.
Actual Result	Both links were displayed correctly.
Status	Pass

TC-04 Output:

Useful Links

Visit the official [CollegeWebsite](#) for more details.

For external research, you can search on [Google](#).

Test Case	TC-05
Test / Input	Check the Student Details table

Expected Result	The table should display Roll No, Name, Branch, and Email details for the students.
Actual Result	The Student Details table was displayed correctly with the required columns and student information.
Status	Pass

TC-05 Output:

Student Details

Roll No	Name	Branch	Email
100	Raj	CSE	raj@gmail.com
101	Priya	CSM	priya@gmail.com

Test Case	TC-06
Test / Input	Check the College Logo and Campus Image
Expected Result	The college logo and campus image should be displayed correctly on the webpage.
Actual Result	Both images were rendered correctly.
Status	Pass

TC-06 Output:

College Logo



Campus Image



Test Case	TC-08
Test / Input	Check the Audio and Video elements
Expected Result	The webpage should display functional audio and video media controls.
Actual Result	Both audio and video elements were displayed correctly with media controls.
Status	Pass

TC-08 Output:

Audio



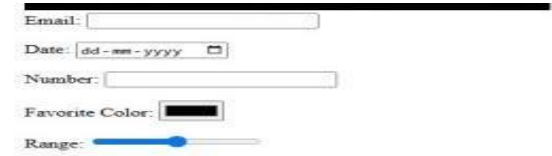
Video



Test Case	TC-09
Test / Input	Check the additional input controls
Expected Result	Email, Date, Number, Favorite Color, and Range input controls should be displayed correctly.

Actual Result	All additional input controls were displayed correctly.
Status	Pass

TC-09 Output:



The screenshot shows a registration form with the following controls: an Email text input field, a Date input field with a calendar icon, a Number text input field, a Favorite Color color picker, and a Range slider.

Test Case	TC-10
Test / Input	Check the Submit and Reset buttons in the registration form
Expected Result	Submit and Reset buttons should be displayed and available for form interaction.
Actual Result	Both Submit and Reset buttons were displayed correctly.
Status	Pass

TC-10 Output:



The screenshot shows two buttons, 'Submit' and 'Reset', side-by-side.

9. OUTPUT

Output 1: College Information Portal

College Information Portal

About Our Institute

Welcome to the official Information Portal

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Academic Excellence

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Check out our award-winning formula : $E=mc^2$. Alternatively, look at our chemistry lab schedules for H_2O solutions.

Available Courses

- Computer Science Engineering
- Data Science
- Artificial Intelligence

1. Fill out the online application form.
2. Submit your Academic transcript.
3. Attend the counseling session.

Useful Links

Visit the official [CollegeWebsite](#) for more details.

For external research, you can search on [Google](#).

Student Details

Roll No	Name	Branch	Email
100	Raj	CSE	raj@gmail.com
101	Priya	CSM	priya@gmail.com

College Logo



Campus Image



Student Registration Form

Name:

Email:

Password:

Gender: ☐ Male ☐ Female

Hobbies: ☐ Sports ☐ Reading ☐ Music

Branch:

Comments:

College Information Portal

[Home](#) [About](#) [Contact](#)

About College

ANITS college provides quality education.

Audio



Video



Email:

Date:

Number:

Favorite Color:

Range:

Copyright 2026 ANITS College

Figure 1: Output of the College Information Portal webpage showing headings, paragraphs, formatting, lists, hyperlinks, table, images, and form.

Supporting Files Used

The following files are used with the HTML page:

- logo.jpg — College logo
- campus.jpg — Campus image
- audio.mp3 — Audio file
- video.mp4 — Video file logo.jpg



campus.jpg



10. INTERVIEW QUESTIONS

Q1. What is the purpose of semantic HTML5 elements?

Answer:

Semantic HTML5 elements provide meaningful information about the structure and purpose of different sections of a web page. Elements such as `<header>`, `<nav>`, `<section>`, `<article>`, and `<footer>` make the HTML code easier to understand, maintain, and interpret by browsers, search engines, and assistive technologies.

Code Snippet:

```
<header>
  <h1>College Information Portal</h1>
</header>

<nav>
  <a href="#">Home</a>
  <a href="#">About</a>
  <a href="#">Contact</a>
</nav>

<section>
  <article>
    <h2>About College</h2>
    <p>ANITS college provides quality education.</p>
  </article>
</section>
```

```
<footer>
  <p>Copyright 2026 ANITS College</p>
</footer>
```

Q2. Compare Ordered and Unordered Lists.

Answer:

An **ordered list** () displays items in a specific sequence, usually using numbers or letters. An **unordered list** () displays items without indicating a particular order, usually using bullet points.

Code Snippet:

```
<!-- Unordered List -->
<ul>
  <li>Computer Science Engineering</li>
  <li>Data Science</li>
  <li>Artificial Intelligence</li>
</ul>

<!-- Ordered List -->
<ol>
  <li>Fill out the online application form.</li>
  <li>Submit your Academic transcript.</li>
  <li>Attend the counseling session.</li>
</ol>
```

Q3. What happens if an image path is incorrect?

Answer:

If the image path is incorrect, the browser cannot find and display the image. Instead, the image will appear as broken or missing. The alternative text specified using the alt attribute may be displayed.

Code Snippet:

```
<h2>College Logo</h2>


<h2>Campus Image</h2>

```

Q4. Differentiate Radio Buttons and Checkboxes.

Answer:

Radio buttons are generally used when the user must select **one option from a group of options**. Checkboxes are used when the user can select **multiple independent options**.

Code Snippet:

Gender:

```
<input type="radio" name="Gender">Male  
<input type="radio" name="Gender">Female
```

```
<br><br>
```

Hobbies:

```
<input type="checkbox">Sports  
<input type="checkbox">Reading  
<input type="checkbox">Music
```

Q5. Explain the advantages of HTML5 input types over traditional text boxes.

Answer:

HTML5 input types provide specialized controls and built-in validation. For example, email validates email-format input, date provides a date selector, number supports numeric values, color provides a color picker, and range provides a slider. They improve usability and reduce the need for additional validation code.

Code Snippet:

```
<form>
```

```
  Email: <input type="email"><br><br>
```

```
  Date: <input type="date"><br><br>
```

```
  Number: <input type="number"><br><br>
```

```
  Favorite Color: <input type="color"><br><br>
```

```
  Range: <input type="range"><br><br>
```

```
</form>
```

11. RESULT

Thus, a **College Information Portal** was successfully developed using **HTML and HTML5**. Basic HTML elements, text formatting, lists, hyperlinks, tables, images, forms, semantic elements, multimedia, and HTML5 input types were implemented and verified successfully.